Gopala Krishna Akkineni

DevOps Engineer | Bloomington, IL | +1 (217) 306-7105 | gopalakrishnaakkineni.dev@gmail.com | linkedin.com/in/gopala-krishna-devops

Professional Summary

DevOps Engineer with over 4 years of progressive experience in automating, optimizing, and securing multi-cloud infrastructures across AWS, GCP, and Azure. Expert in building scalable CI/CD pipelines, orchestrating containerized workloads with Kubernetes, and implementing Infrastructure as Code (IaC) practices with Terraform and CloudFormation. Possess hands-on expertise in supporting AI/ML workflows by deploying GPU-enabled environments, securing secrets, and integrating monitoring/alerting solutions to maintain high availability. Known for improving deployment speed by up to 40%, reducing infrastructure costs by 15%, and ensuring 99.9% system uptime. Highly collaborative team player with proven ability to work in Agile/Scrum environments alongside developers, QA, and ML engineers to deliver innovative, secure, and efficient cloud solutions.

Core Skills

Cloud Platforms: AWS (EC2, EKS, Lambda, RDS, S3, IAM, Route 53), GCP (GKE, BigQuery, Cloud Functions, Cloud Storage), Azure (AKS, Key Vault, Blob Storage).
Infrastructure as Code: Terraform, CloudFormation, Deployment Manager, Bicep.
CI/CD & GitOps: Jenkins, GitHub Actions, Azure DevOps, Buildkite, ArgoCD, FluxCD.
Containers & Orchestration: Docker, Kubernetes (EKS, GKE, AKS), Helm.
Monitoring & Logging: ELK Stack, Prometheus, Grafana, CloudWatch.
Security: Vault, AWS Secrets Manager, GCP Secret Manager, SonarQube, Snyk, Trivy.
Programming & Automation: Python, Bash, Shell scripting for automation and ML pipelines.
Collaboration: Git, GitHub/GitLab, JIRA, Rally, Agile/Scrum methodologies.

Professional Experience

DevOps Engineer (AWS/GCP), State Farm, Bloomington, IL — Mar 2025 to Present

- Led the design and implementation of a scalable, secure, and automated multi-cloud infrastructure to support end-to-end AI/ML model training, deployment, and monitoring across AWS and GCP.
- Developed, optimized, and maintained CI/CD pipelines using GitHub Actions, Jenkins, and Buildkite to accelerate code delivery, reducing release cycles by 40%.
- Provisioned, secured, and managed cloud infrastructure using Terraform and CloudFormation, achieving high reliability with 99.9% uptime.
- Managed Kubernetes (EKS/GKE) clusters, configured Helm charts, integrated GitOps tools like ArgoCD/FluxCD, and implemented auto-scaling policies for high-traffic applications.
- Orchestrated GPU/TPU-enabled clusters for deep learning workloads, cutting training and inference times by up to 30%.
- Implemented blue-green and canary deployment strategies to ensure zero-downtime deployments, minimizing service interruptions by 95%.
- Integrated ML lifecycle management tools such as SageMaker, Bedrock, and Hugging Face into DevOps pipelines, streamlining model deployment.
- Designed and implemented secure secret management using Vault, AWS Secrets Manager, and GCP Secret Manager, strengthening compliance and reducing security risks.
- Established comprehensive observability with Prometheus, Grafana, CloudWatch, and ELK Stack, enabling proactive detection of anomalies and reducing downtime by 25%.
- Automated resource monitoring and scaling for GPU-based ML workloads, reducing costs by 15% while maintaining performance.
- Collaborated with data scientists and ML engineers to deploy LLM-based applications into production environments.
- Performed incident response and root cause analysis, documenting lessons learned and implementing fixes for long-term system stability.

Cloud/DevOps Engineer, Amistad Technologies, Hyderabad, India May 2022 to Jun 2023

- Provided L1/L2 support for AWS services (EC2, S3, IAM) and resolved networking-related incidents, achieving a 95% SLA adherence rate.
- Assisted in configuring and troubleshooting CI/CD pipelines in Jenkins, GitHub Actions, and Azure DevOps, improving deployment reliability.
- Automated repetitive tasks such as backups, log archiving, and IAM provisioning with Bash and Python scripts integrated with Lambda functions.
- Developed reusable Terraform modules for S3 buckets, IAM roles, and EC2 instances, cutting manual provisioning by 50%.
- Set up CloudWatch alarms, dashboards, and log filters to enhance proactive system monitoring and troubleshooting.
- Participated in weekly incident review sessions and root cause analyses, contributing actionable insights for future prevention.
- Supported disaster recovery exercises by validating backup restorations and DNS failover testing, enhancing resilience.
- Collaborated with DevOps and InfoSec teams to rotate IAM credentials, enforce least-privilege policies, and manage secrets securely.
- Assisted in cost optimization by identifying underutilized AWS resources and implementing automated shutdowns, reducing costs by 10%.
- Improved deployment success rates by validating infrastructure readiness during release windows, cutting failure rates by 20%.

Cloud/DevOps Engineer, Taal Tech, Bangalore, India Mar 2021 to Feb 2022

- Supported the deployment and maintenance of multi-cloud environments across AWS, Azure, and GCP, assisting in provisioning EC2 instances, virtual machines, and storage buckets.
- Configured and managed IAM roles, security groups, and key rotations, ensuring secure access control and compliance with internal standards.
- Developed and executed Shell and Python scripts to automate health checks, log collection, and cleanup processes, reducing manual monitoring tasks.
- Contributed to Terraform and CloudFormation templates for consistent environment provisioning and version-controlled infrastructure.
- Assisted in CI/CD pipeline setup using Jenkins and GitHub Actions, automating build, test, and deployment steps for internal applications.
- Monitored system performance and resource utilization using AWS CloudWatch and Azure Monitor, identifying and escalating potential performance issues.
- Supported Kubernetes environment troubleshooting, addressing pod restarts, configuration errors, and resource allocation adjustments under supervision.
- Collaborated with cross-functional teams during Agile sprints, documenting processes and contributing to onboarding materials and runbooks.

Linux Administrator, Ignitarium, Kochi, India Jul 2020 to Feb 2021

- Supported development teams in setting up and troubleshooting Linux-based servers to ensure application performance.
- Monitored server health and system performance, reporting issues and escalating critical failures.
- Wrote shell scripts to automate tasks like user provisioning, file management, and process monitoring, reducing manual intervention.
- Configured and maintained user permissions, cron jobs, and log rotation policies to ensure secure and efficient Linux server operations.
- Performed routine system patching, kernel updates, and security hardening to maintain compliance with organizational standards.
- Helped configure log monitoring and scheduled backups using simple shell scripts to ensure system stability and data protection.

Project Highlights

- **AI/ML Infrastructure Automation:** Designed Terraform-based pipelines to provision GPU clusters for ML training, reducing costs by 20% while accelerating deployment.
- **GitOps Deployment:** Implemented ArgoCD workflows that enabled fully automated rollbacks, reducing downtime and recovery from hours to minutes.
- **Security Automation:** Integrated Snyk, Trivy, and SonarQube into CI/CD workflows, improving code compliance and reducing vulnerabilities by 35%.
- **Monitoring & Cost Optimization:** Built Grafana dashboards and predictive alerts, enhancing observability and reducing downtime by 25%.
- **Multi-Cloud Resiliency:** Architected hybrid workloads on AWS and GCP, enabling cross-cloud failover strategies for business continuity.

Education

- Master’s in Management Information Systems
- Bachelor of Engineering in Electronics and Communication
- Diploma in Electronics and Communication Engineering